

ASSIGNMENT #5

Course: CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2020
Professor: Dr. Michael Caracotsios
Office: EIB 244
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Your name: _____

Due Date: Saturday April 04 by 12:00 noon

Number of points.....100

Your score.....

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Scan your Assignment and create a PDF file
- PDF File Name: Assignment5_XY.PDF
 - X: First letter of your first name
 - Y: Your last name
 - Example: Assignment5_MCaracotsios.PDF
- Email your PDF file to: Suzane Carneiro svieir2@uic.edu by the due date (Saturday April 04) and time (12:00 noon)

Note: All problems are worth 100 points. Your final score will be the average value

Problem 5A: Calculate the fugacity of water at $\{T = 250\text{ }^{\circ}\text{C} \text{ and } P = 70\text{ bar}\}$ with following methods:

1. Steam Tables.
2. Assuming the water follows the Peng-Robinson Equation of State both in the gas and liquid phases.
3. Assuming the water follows the Peng-Robinson Equation of State only in the gas phase.

Note: For methods 2 and 3 you are not allowed to use the steam tables.

Problem 5B: Pure propane is maintained at $\{T = 50\text{ }^{\circ}\text{C} \text{ and } P = 5\text{ bar}\}$

1. Verify whether propane is in the liquid or gas phase or whether there are two phases present. Assume propane follows the van der Waals equation of state
2. Calculate the numerical value of the derivative $\left(\frac{\partial T}{\partial P}\right)_s$ assuming propane is ideal gas
3. Calculate the numerical value of the derivative $\left(\frac{\partial T}{\partial P}\right)_s$ by using first order numerical differencing and assuming that propane follows the van der Waals equation of state
4. Develop an expression for the derivative $\left(\frac{\partial T}{\partial P}\right)_s$ using the van der Waals equation of state and calculate its numerical value.
5. Prepare a table that shows the results of the three methods.

Note: First order difference formula for the approximation of the first order derivative of a continuous function is given by:

$$\frac{df}{dx} \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$